

Mealy Machine Examples

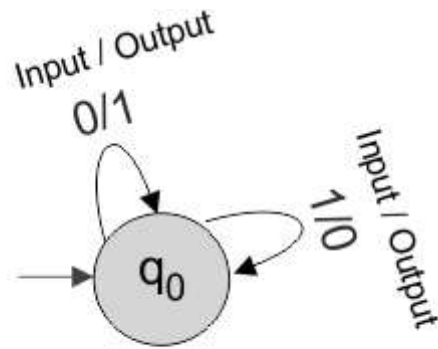
As we already covered the topic of Automata Mealy Machine. In this lecture, we will see some Examples of Automata Mealy machine.

Example 1: Mealy Machine for the first complement

Design a Mealy machine to generate the first complement of any given binary string.

Solution

According to the first complement, if the given input is “1”, the output will be “0” and vice versa. Hence, the following is the **Mealy Machine** of the given question



1st Complement in Mealy Machine

A Mealy Machine transition table for given example is given below

Current State	Input "0"	Input "1"
q_0	$q_0, 1$	$q_0, 0$

Mealy Transition Table
First Complement Transition Table

For instance, take one binary number, 11100, then look at the following table.

Input	1	1	1	0	0
State	q_0	q_0	q_0	q_0	q_0
Output	0	0	0	1	1

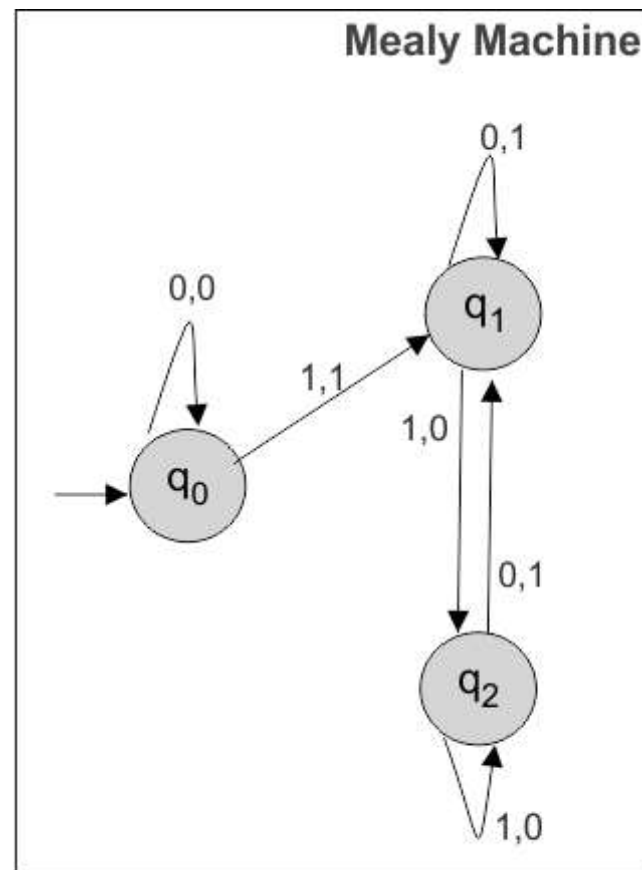
Thus, we get 00011, which is the 1's complement of 11100,

Example 2: Mealy Machine for the 2nd complement

Design a Mealy machine to generate the second complement of any given binary input.

Solution

The following is the **Mealy Machine** of the given example



Transition diagram of **2's Complement**

A Mealy Machine transition table for the given example is given below

Current State	Input "0"	Input "1"
q_0	$q_0, 0$	$q_1, 1$
q_1	$q_1, 1$	$q_2, 0$
q_2	$q_1, 1$	$q_2, 0$

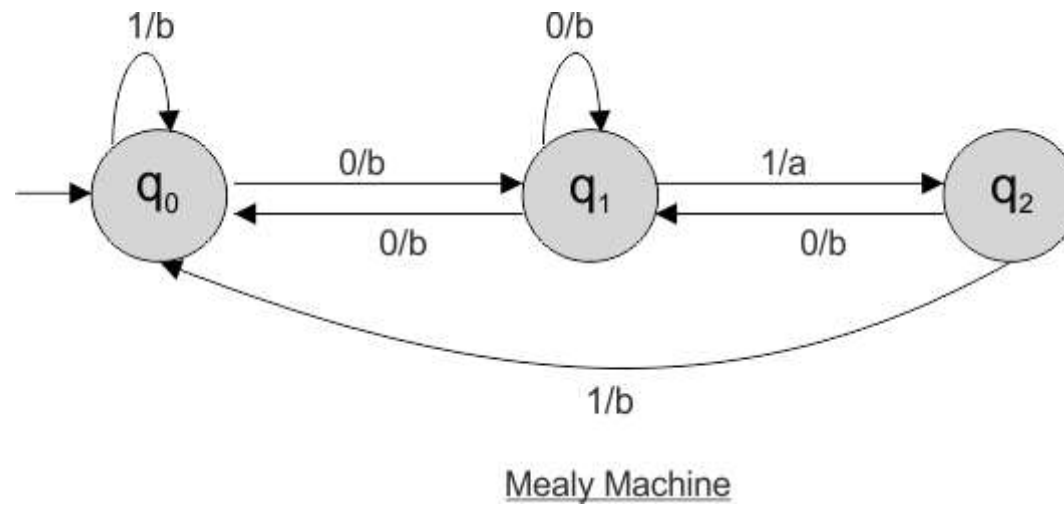
Mealy Transition Table
Transition Table of 2's Complement

Example 03 – Mealy Machine to print “a” whenever the sequence “01” is encountered

Design a Mealy Machine that prints “a” whenever the sequence “01” is encountered in any input binary string.

Solution

The requirement is that whenever the “01” appears in the string, the output must end with “a”. The following is the desired **Mealy Machine**.



A Mealy Machine transition table for the given example is given below

Current State	Input "0"	Input "1"
q_0	q_1, b	q_0, b
q_1	q_1, b	q_2, a
q_2	q_1, b	q_0, b

Mealy Transition Table

to prints "a" whenever the sequence "01" occur

For instance, take one binary number, 10111001, and then look at the following table.

Input	1	0	1	1	1	0	0	1
State	q0	q1	q2	q0	q0	q1	q1	q2
Output	b	b	a	b	b	b	b	a

As shown in the above table, whenever the “01” occurs in the input string, the output will be “a”.

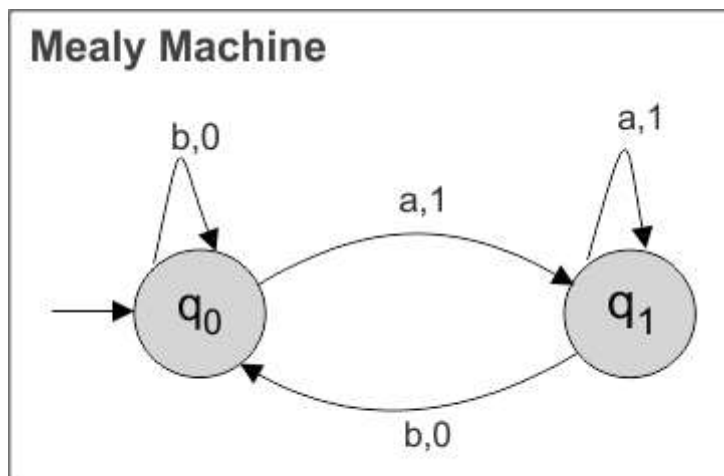
Example 04 – Mealy Machine to count the occurrences of “a’s”

Construct a Mealy Machine that takes all the strings of a’s and b’s as input and counts the number of a’s in the input string in terms of 1. Where

input $\Sigma = \{a,b\}$, and output $\Delta = \{0,1\}$

Solution

Mealy automata machine, for the given example to counts the number of a’s in the input string in terms of 1, is given below



Transition Diagram to counts the **number of a's**

The Transition Table for the given example (counts the number of a's in the input string in terms of 1) is given below

Current State	Input "a"	Input "b"
q_0	$q_1, 1$	$q_0, 0$
q_1	$q_1, 1$	$q_0, 0$

Mealy Transition Table
Counts the **number of a's**

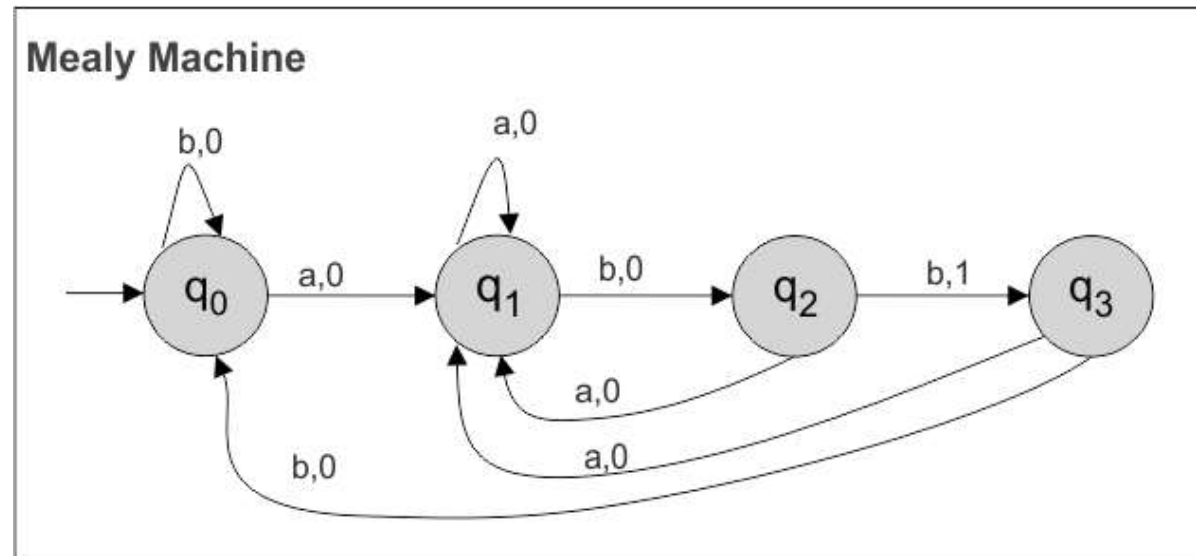
Example 05 – Mealy Machine to count occurrences of “abb”

Design a Mealy machine that counts the occurrences of the sequence “abb” In any input string over $\{a,b\}$

input $\Sigma = \{a,b\}$, output $\Delta = \{0,1\}$

Solution

Mealy automata machine, to count the occurrences of the sequence “abb” In any input string , is given below



Transition Diagram to Counts the **occurrence of “abb”**

The Mealy Machine transition table provides a count of the occurrences of the sequence “abb” in any given input string

Current State	Input "a"	Input "b"
q_0	$q_1, 0$	$q_0, 0$
q_1	$q_1, 0$	$q_2, 0$
q_2	$q_1, 0$	$q_3, 1$
q_3	$q_1, 0$	$q_0, 0$

Mealy Transition Table
 Transition Diagram to Counts the occurrence of "abb"